

November 20, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics
Engineering

Lesson title:
Bionic energy management

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Energy

The first law of thermodynamics states that a closed system's energy is constant unless energy is transferred in or out by work or heat, and that no energy is lost in transfer. The total inflow of energy into a system must equal the total outflow of energy from the system, plus the change in the energy contained within the system.

The second law of thermodynamics states that the system doing work always loses some energy as waste heat.

Thermodynamic process (3 main kinds):

Change processes are a passage of a thermodynamic system from an initial to a final state of thermodynamic equilibrium.

Cyclic processes defined by a cycle of transfers into and out of a system, a cyclic process is described by the quantities transferred in the several stages of the cycle.

Flow processes defined by flows through a system, a flow process is a steady state of flow into and out of a vessel with definite wall properties.



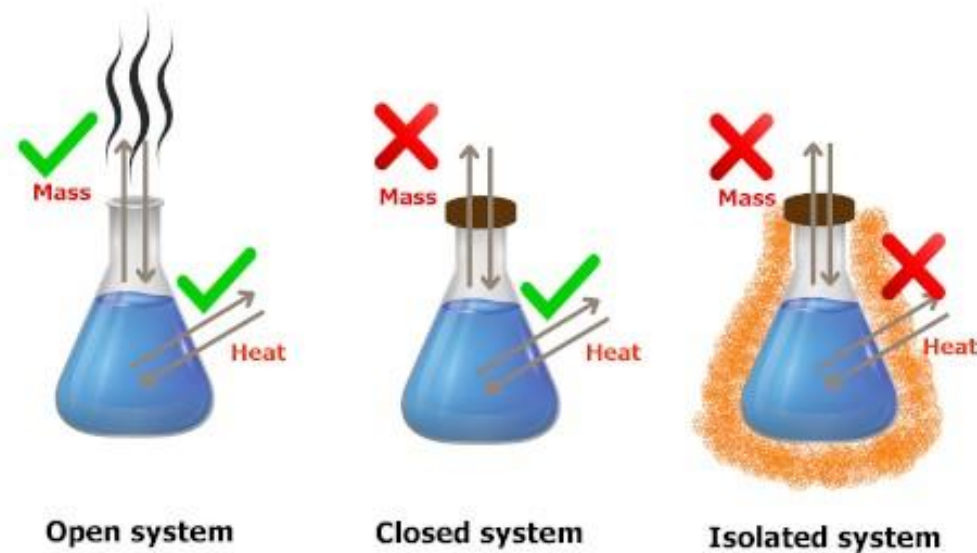
Energy

A thermodynamic system is a body of matter and/or radiation, confined in space by walls, with defined permeabilities (e.g. in terms of all matter, all radiation, and all forces), which separate it from its surroundings.

An **isolated thermodynamic system** can exchange neither matter nor energy with the surrounding environment.

A **closed thermodynamic system** can exchange energy but not matter with the surrounding environment.

An **open thermodynamic system** can exchange both matter and energy with the surrounding environment.



Energy

For **closed systems**, the process of energy transfer is described by the first law

$$\Delta E = W + Q$$

where **E** is the amount of energy transferred, **W** represents the work done on the system, and **Q** represents the heat flow into the system. As a simplification, the heat term, **Q**, is sometimes ignored, especially when the thermal efficiency of the transfer is high.

$$\Delta E = W$$

Open systems can gain or lose energy in association with matter transfer (e.g. by fueling an auto, a system gains in energy without addition of either work or heat). Denoting this energy by **E**, we have

$$\Delta E = W + Q + E.$$



Energy resources

Primary resources, suitable for end use without conversion to another form



Secondary resources, require substantial conversion from a primary source to have the usable form of energy



Energy resources

Renewable, recover their capacity in a time significant by human needs



Non-renewable, significantly depleted by human usage and will not recover their potential significantly during human lifetimes



Fossil fuels



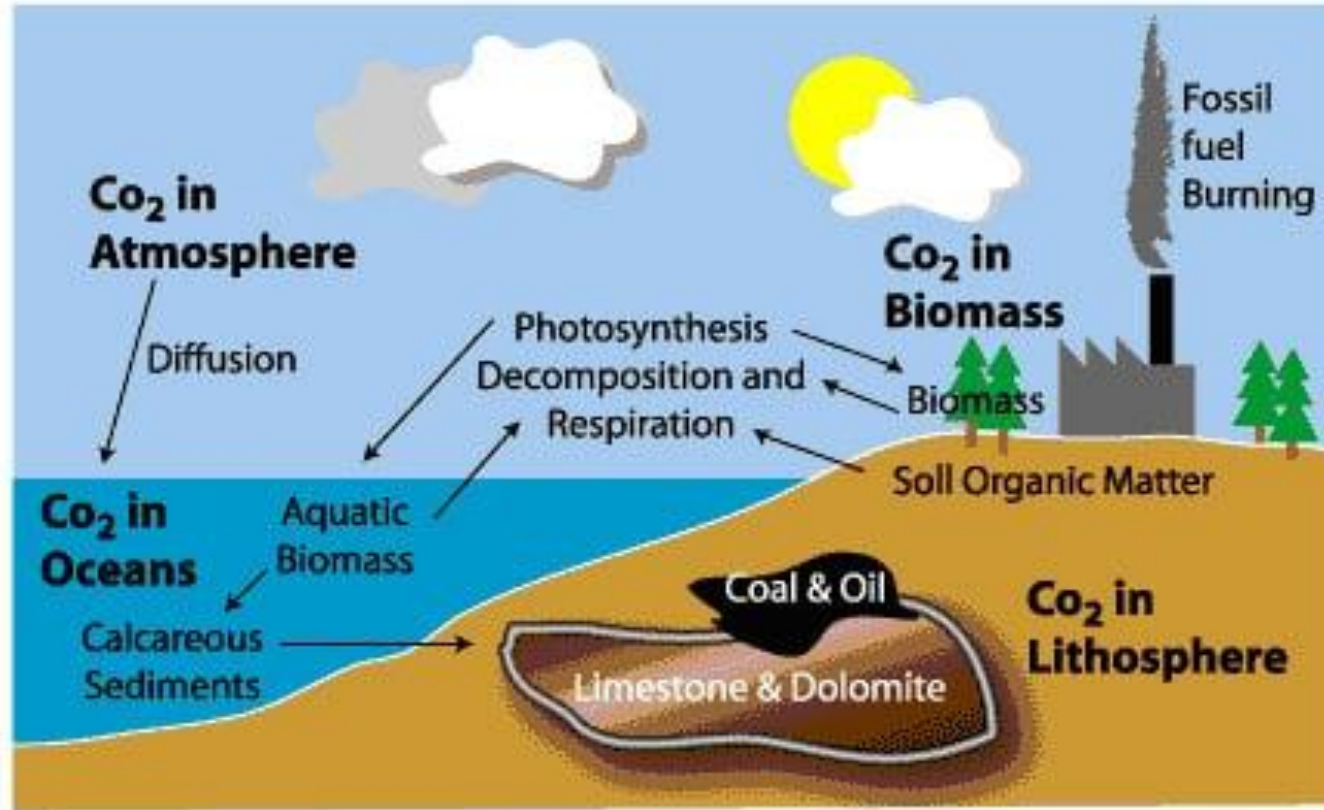
Are the remains of the decomposition of plants and animals and include oil, coal and natural gas

Heat from burning fossil fuels is used for heating or converted to mechanical energy for vehicles, industrial processes, or electrical power generation

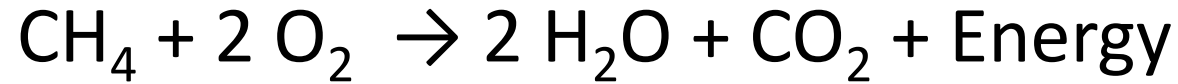


Fossil fuels

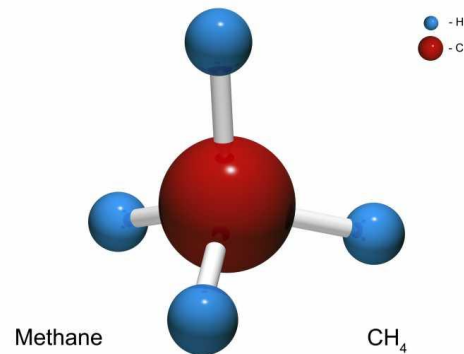
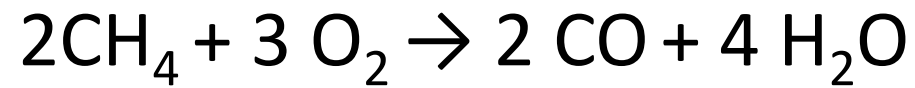
Fossil fuels are part of the **carbon cycle**.
Solar energy, stored over millions of years, is used today.



Fossil fuels



inadequate supply of air (**carbon monoxide** gas and water vapour are formed)



Solar energy

Annual potential of solar energy: **1575–49837 exajoules** (EJ)



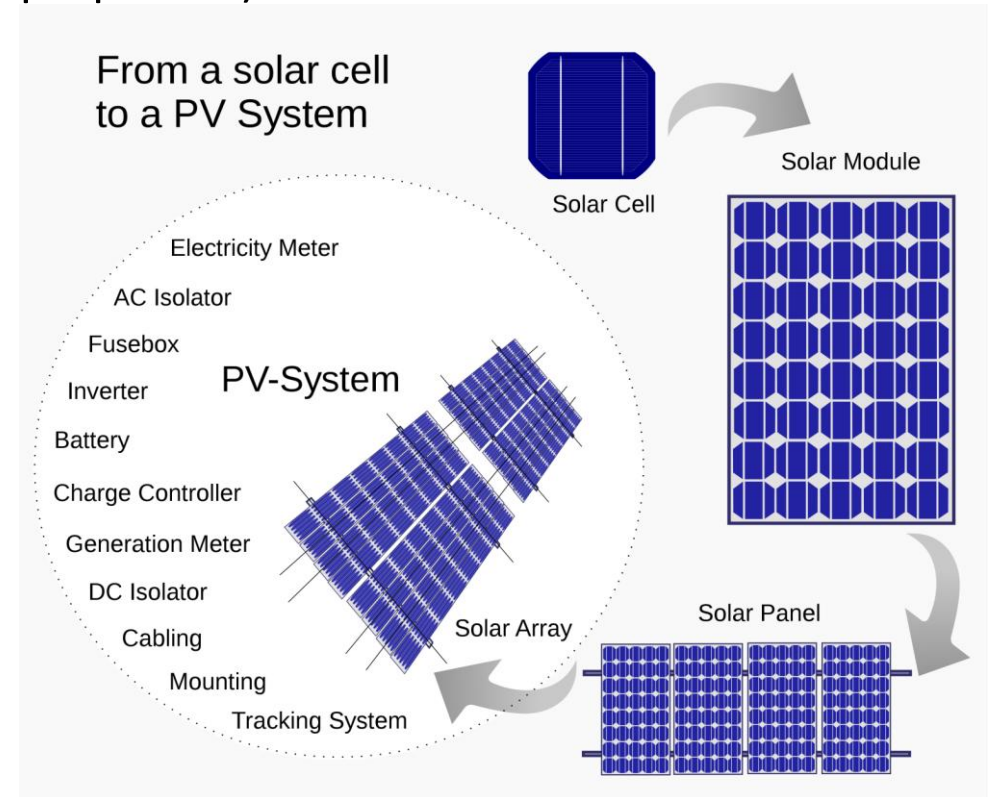
Total world energy consumption: **559.8 EJ**



Solar energy

Photovoltaics generates electrical power by converting solar radiation into direct current electricity using semiconductors (photovoltaic effect, physical and chemical properties)

It employs solar **panels** composed of a number of **solar cells** containing a photovoltaic material (amorphous silicon, monocrystalline silicon, polycrystalline silicon, cadmium telluride, and copper indium gallium selenide/sulfide)

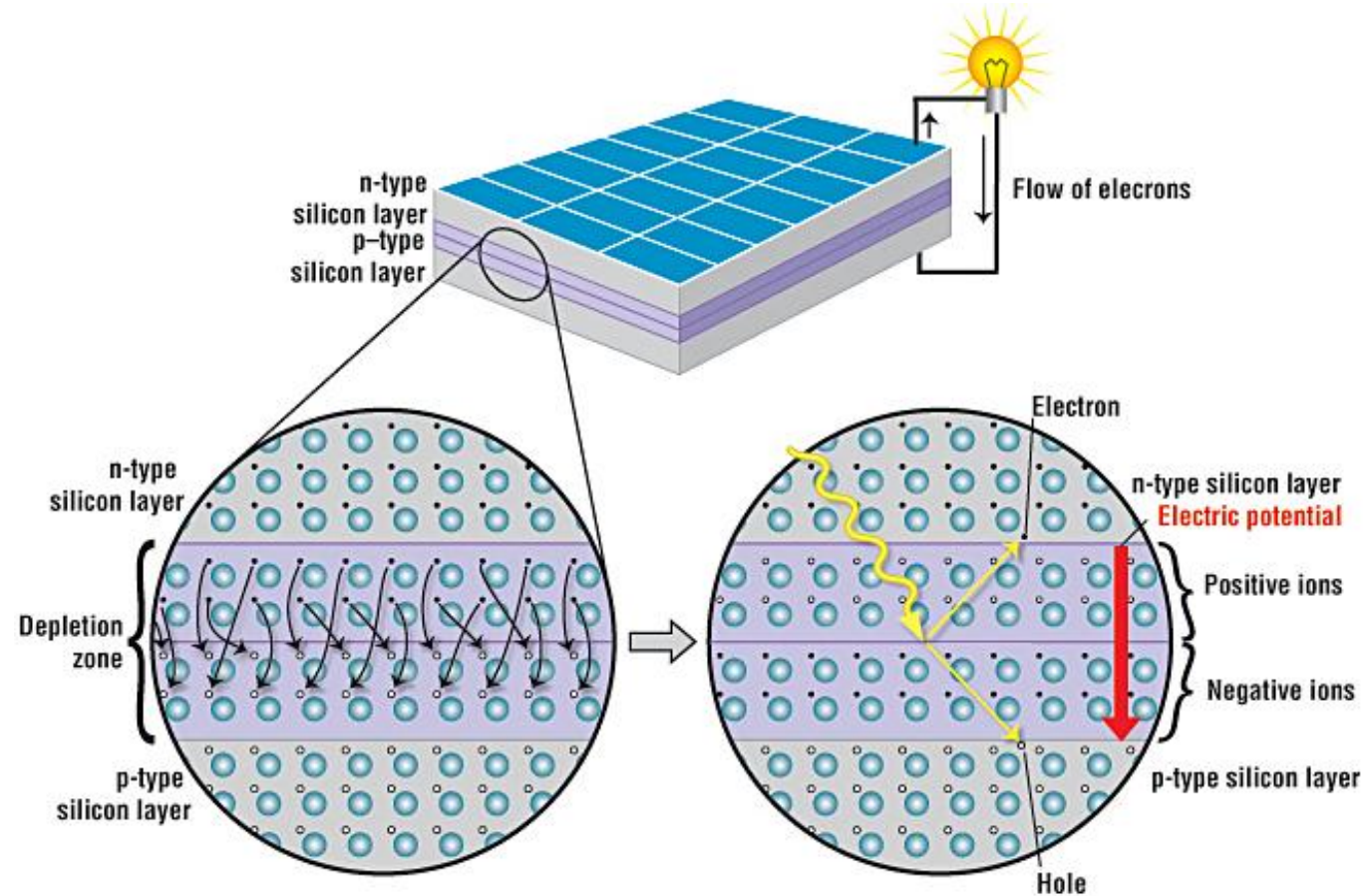


The most used semiconductor material in photovoltaic technology is **silicon**.

One surface of the silicon photovoltaic cell is doped with the insertion of boron atoms (P-type silicon, has electron holes), while the other is doped with small amounts of phosphorus inserted at high temperature (N-type silicon, has an excess number of electrons); by contacting the two P-type and N-type silicon crystals, a P-N junction is obtained. Holes (empty spaces in the orbits) in the P crystal tend to move into the N crystal. Similarly, excess electrons in the N crystal tend to move towards the P crystal.

When photons (energy particles) of sunlight hit the photovoltaic cell, a part of the energy is transformed into heat, a part is reflected, and a last part causes a displacement of the electrons of the atoms in the semiconductor material.

The energy of the photon flow releases a certain number of electrons / holes: the electrons displaced in the atomic structure of the semiconductor start flowing, producing an electric current in the body of the semiconductor.

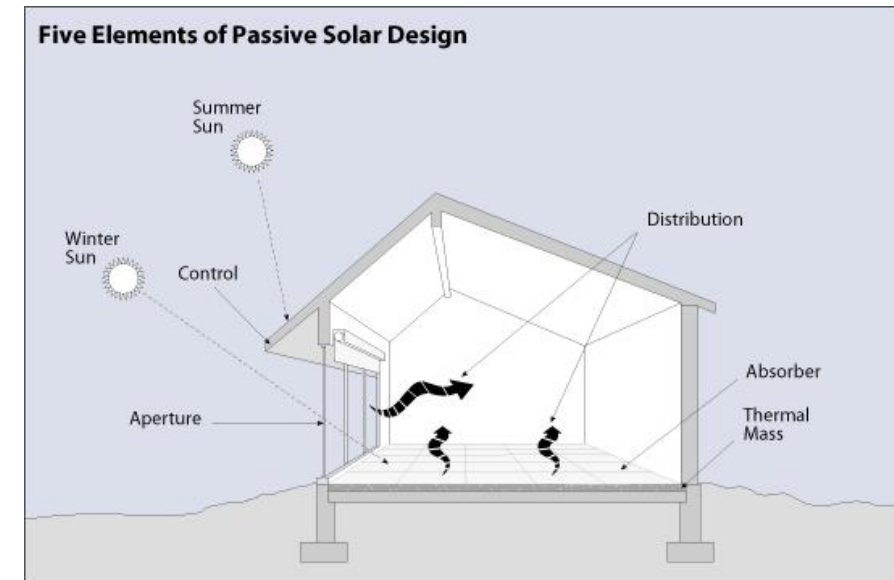


Solar energy

Active solar techniques: the energy is harnessed by the use of **photovoltaic panels** and **solar thermal collectors**.



Passive solar techniques: orienting a building to the Sun, selecting materials with favorable **thermal mass** or light dispersing properties, and designing spaces that naturally circulate air.

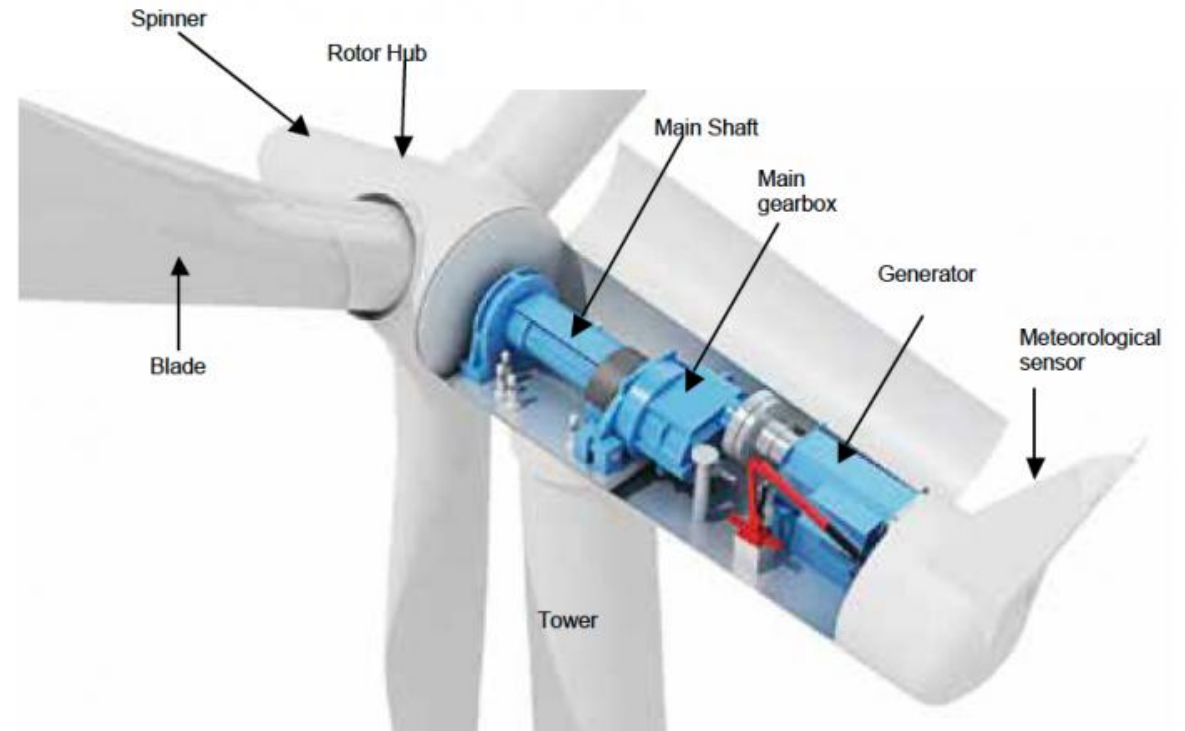


Wind

Differences in **atmospheric pressure** produce air moves from the higher to the lower pressure area

Air flow is converted, through **wind turbines**, to mechanically power generators for electric power.

The maximal achievable extraction of **wind power** by a wind turbine is **59%** of the total theoretical wind power (theoretically).



Wind

Wind energy of the air in motion (Kinetic energy)

$$E = \frac{1}{2} \rho A v^3 t$$

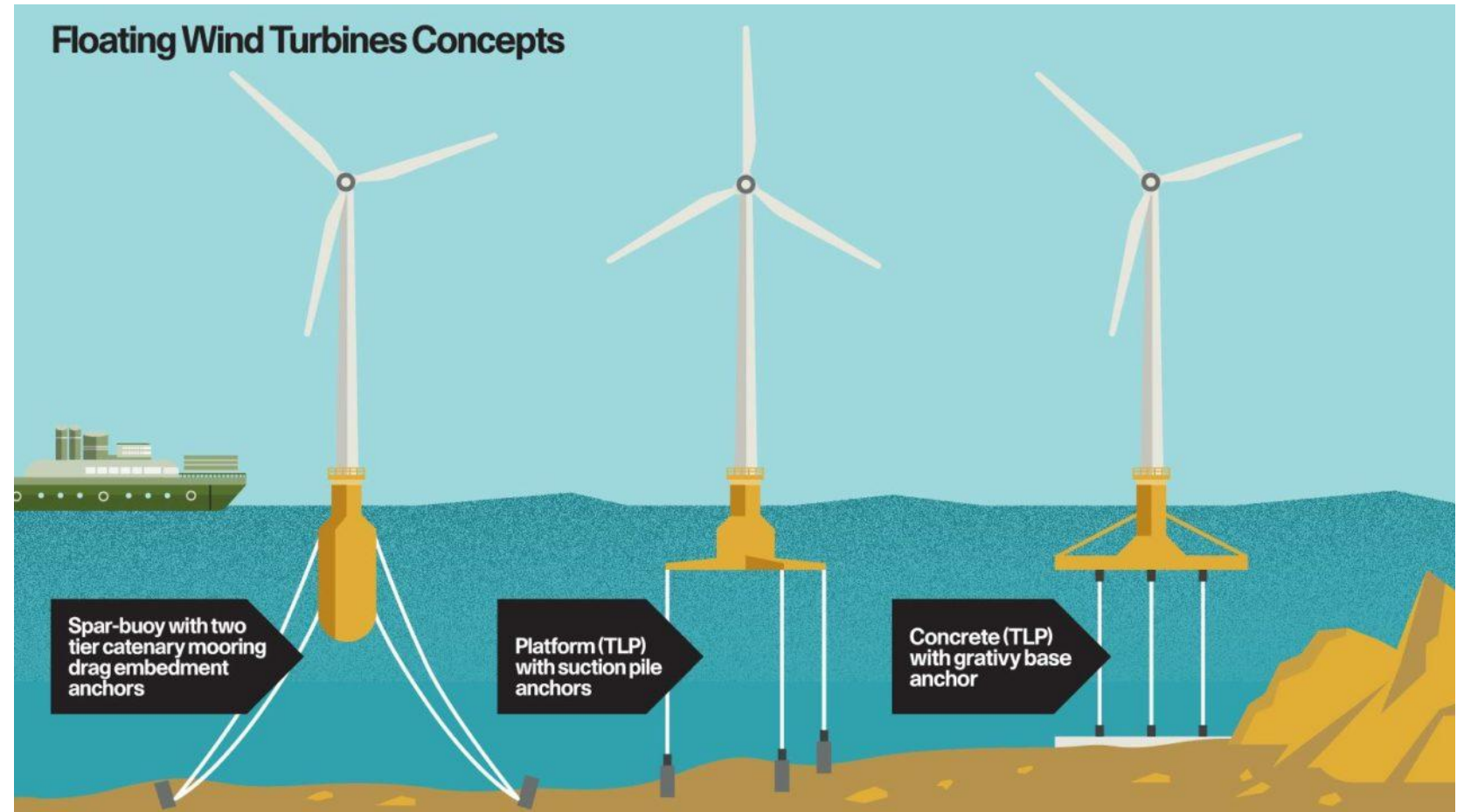
The **wind power** incident on A is:

$$P = \frac{dE}{dT} = \frac{1}{2} A \rho v^3$$



Wind

Offshore wind farms:
harvesting wind energy to
generate electricity in places
where higher wind speeds
are available.



Hydroelectricity

Conventional: the potential energy of dammed water drives a water turbine and generator.

Pumped-storage: this method stores energy in the form of gravitational potential energy of water, pumped from a lower elevation reservoir to a higher elevation.

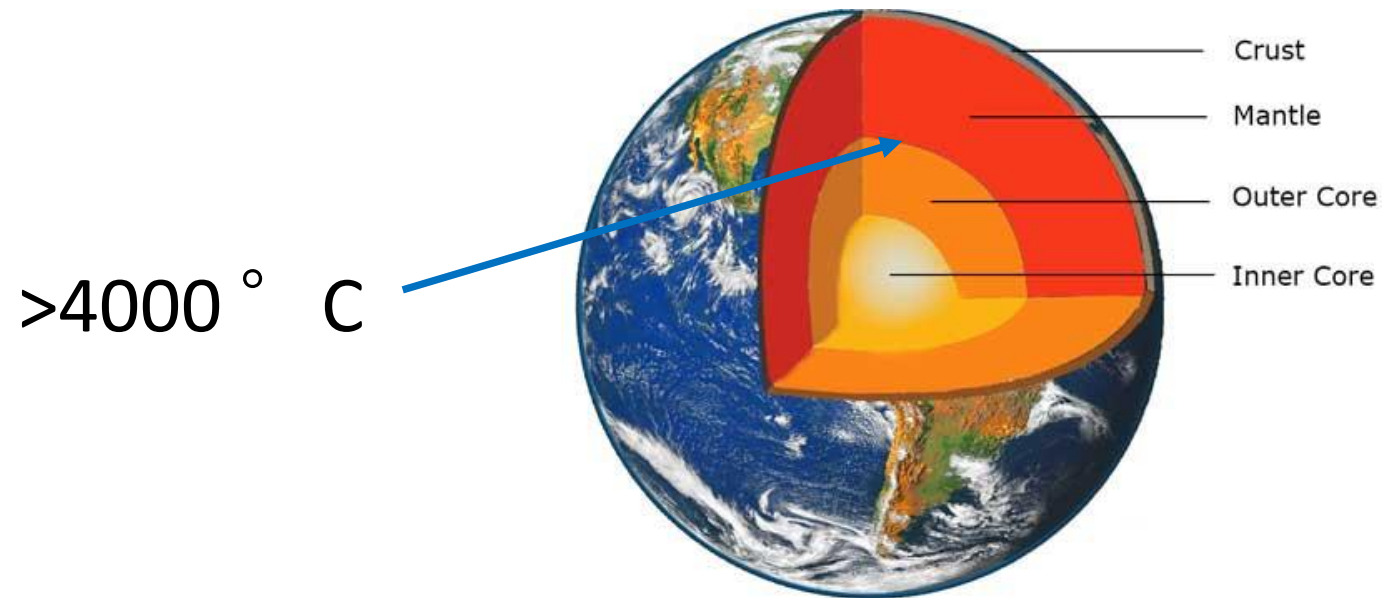
Run-of-the-river: this plant operates as an intermittent energy source. Only the water coming from upstream is available for generation at that moment, and any oversupply must pass unused since these hydroelectric stations have small or no reservoir capacity.

Tidal energy: converts the energy obtained from tides into useful forms of power, mainly electricity



Geothermal

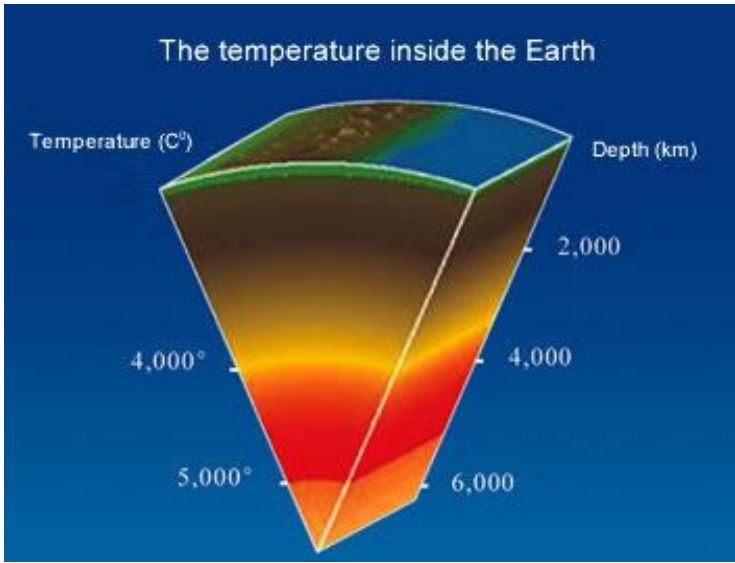
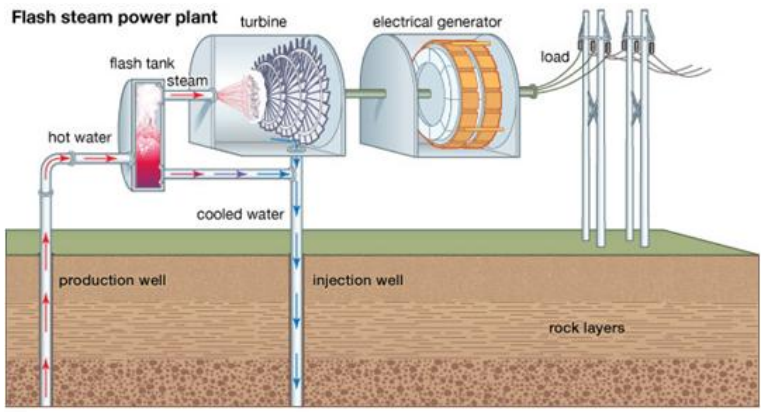
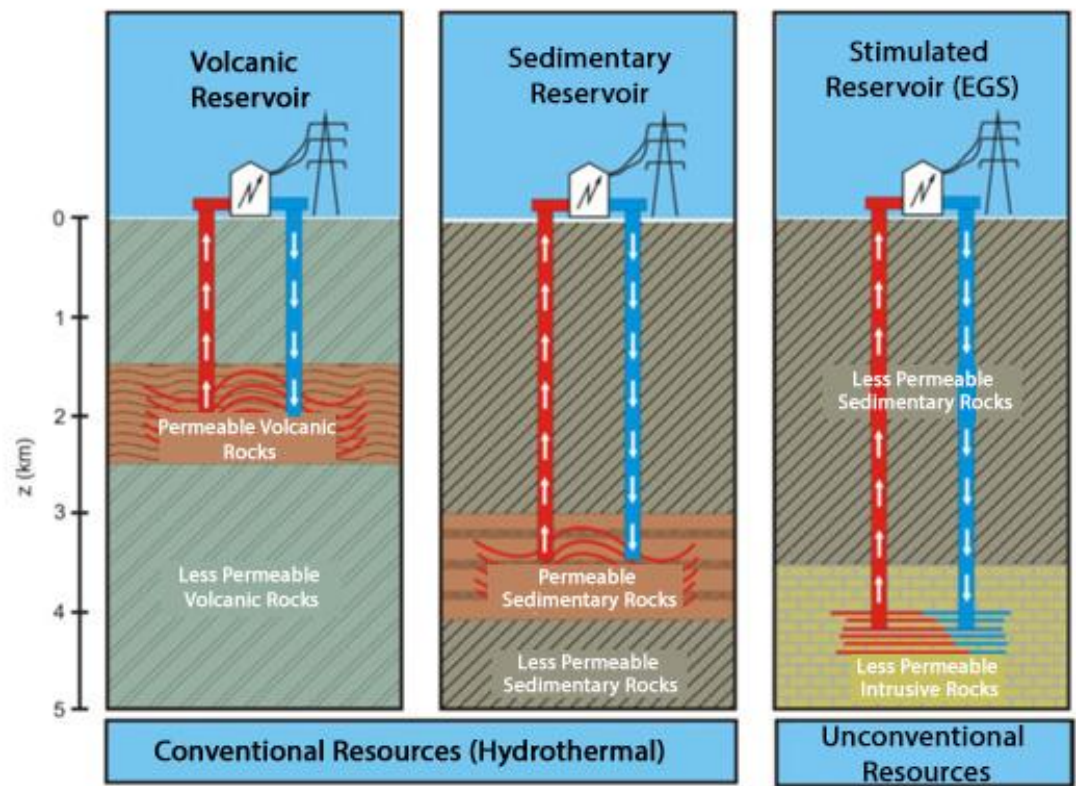
The difference in temperature between the core of the planet and its surface (geothermal gradient) drives a continuous conduction of thermal energy in the form of heat from the core to the surface.



Geothermal energy originates from the original formation of the planet (20%) and from radioactive decay of minerals (80%)

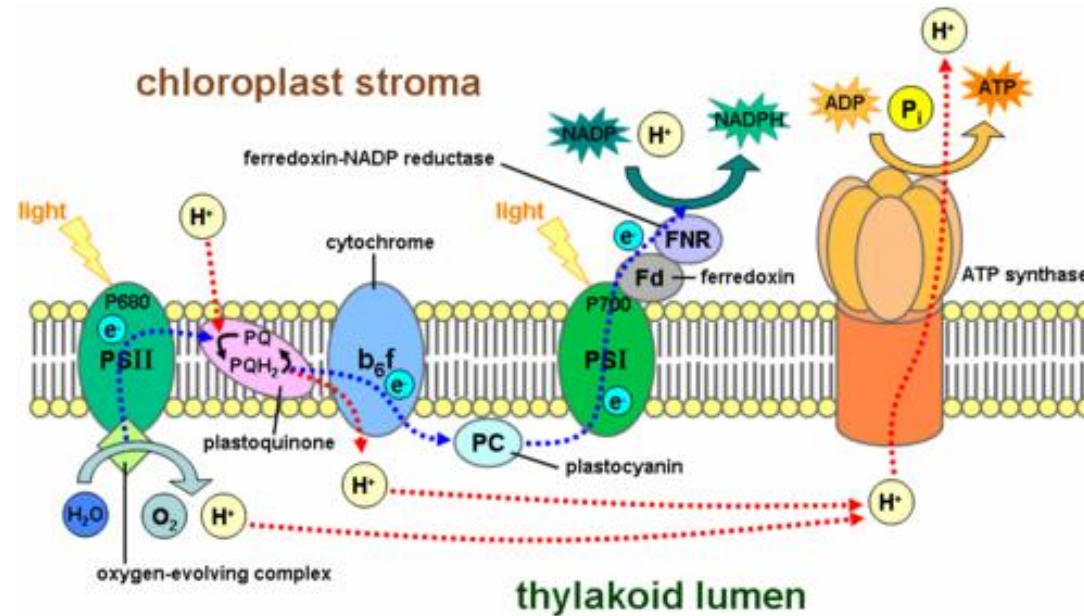
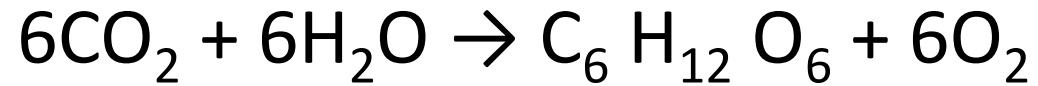


Geothermal



Bioenergy

Biomass deriving from recent carbon fixation occurring in plants and microalgae



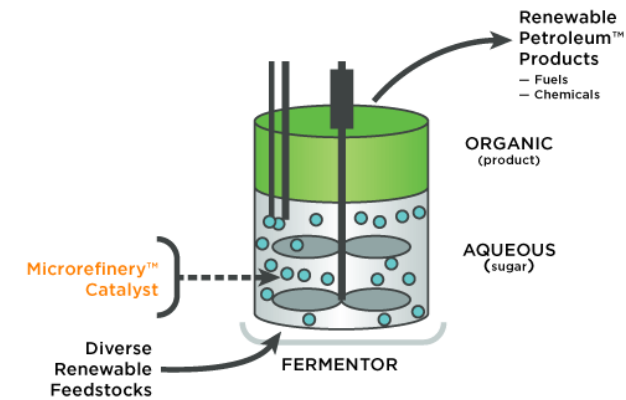
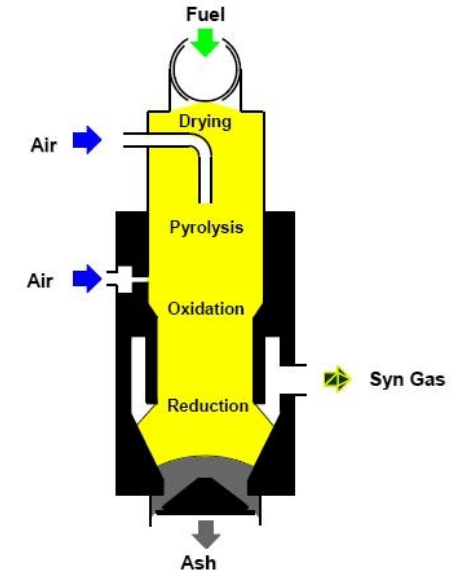
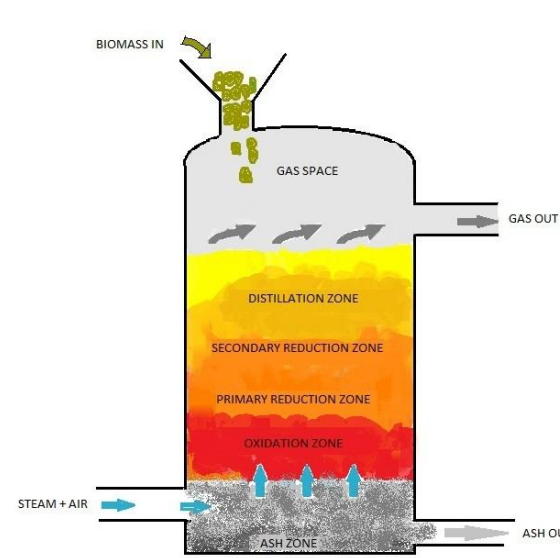
Photoautotrophs, are able to synthesize food directly from carbon dioxide and water using energy from light.



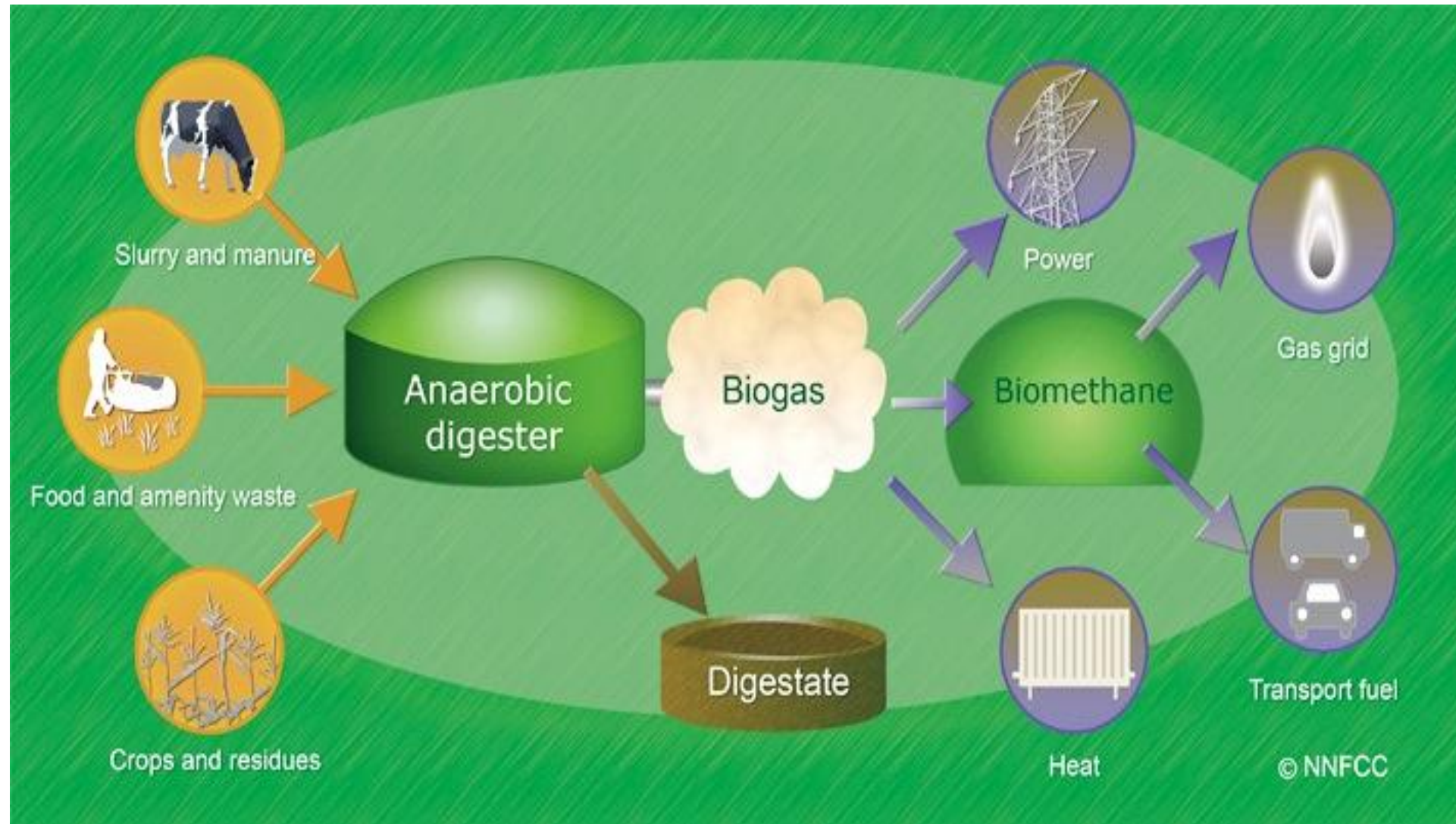
Bioenergy

Biomass conversion:

- Thermal conversions
- Chemical conversion
- Biochemical conversion
- Electrochemical conversion



Bioenergy

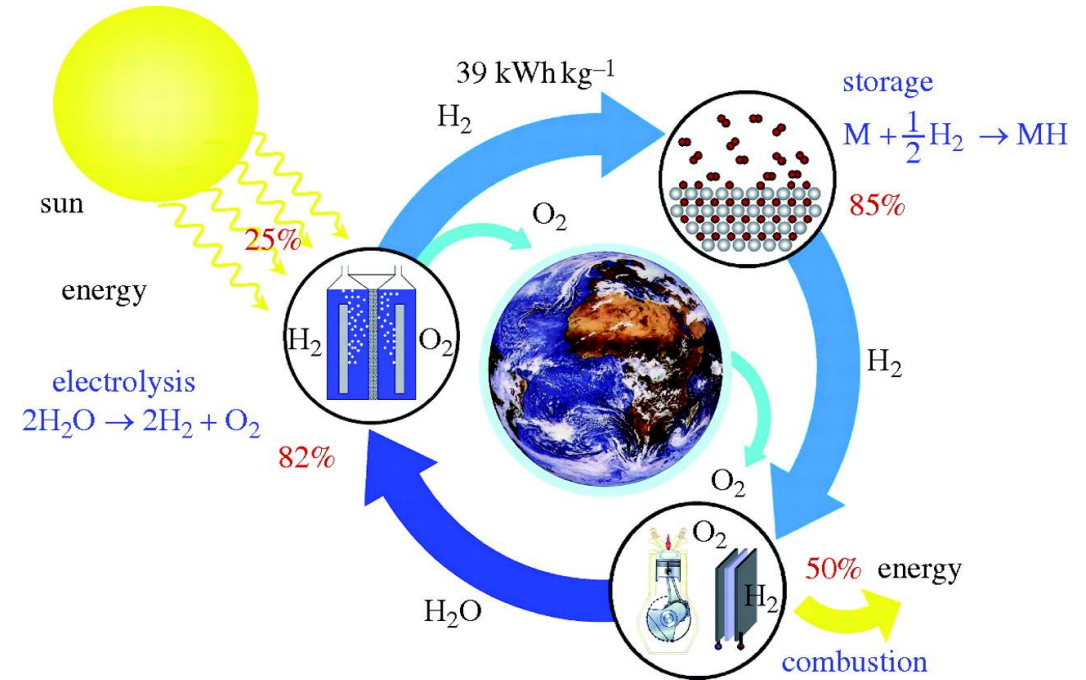


Hydrogen

Hydrogen economy:

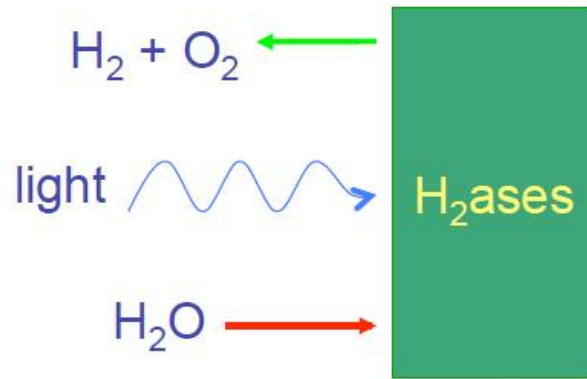
an economy based on solar energy that uses hydrogen to deliver energy.

Hydrogen is an energy carrier, not an energy source.

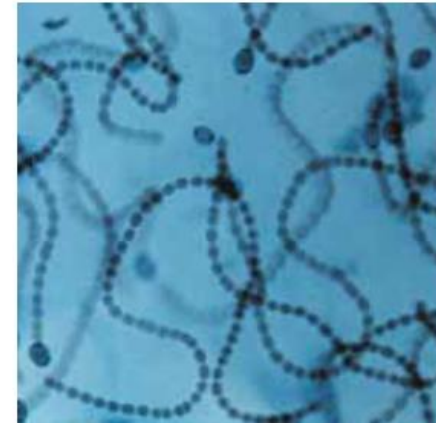
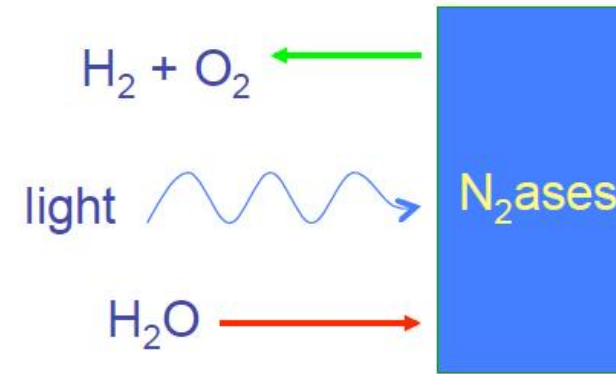


Hydrogen

Photobiological production of hydrogen



green algae

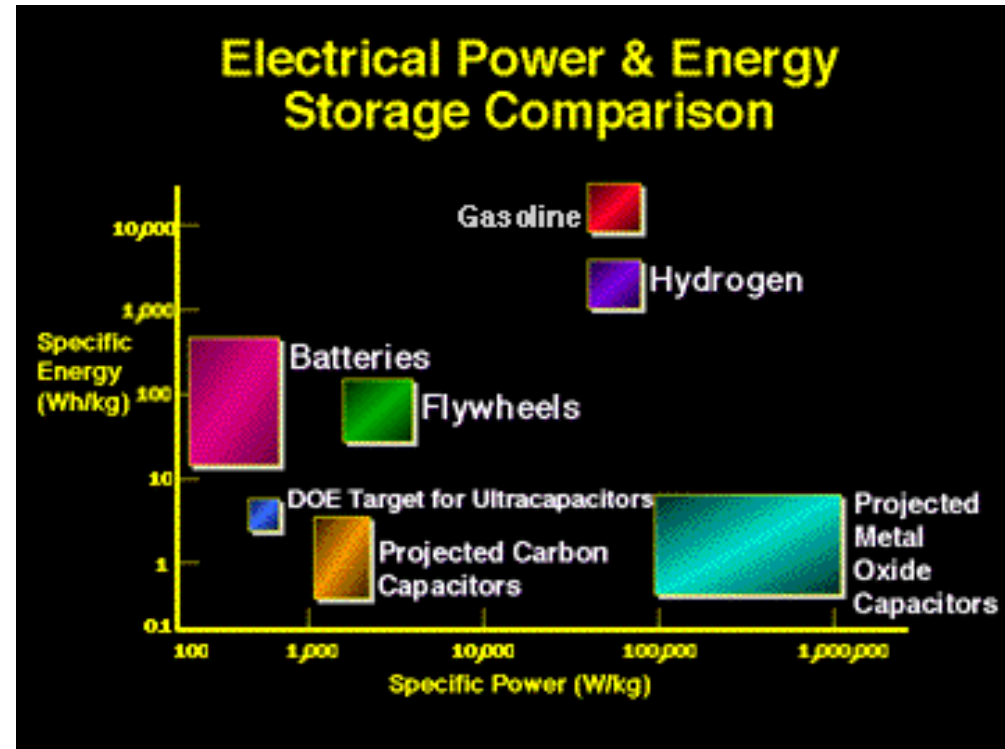


cyanobacteria



Energy Density of Some Materials (KWH/kg)

- Gasoline → 14
- Lead Acid Batteries → 0.04
- Hydrostorage → 0.3 (per cubic meter)
- Flywheel, Steel → 0.05
- Flywheel, Carbon Fiber → 0.2
- Flywheel, Fused Silica → 0.9
- Hydrogen → 38
- Compress Air → 2 (per cubic meter)



Energy density is the amount of energy stored in a given system or region of space per unit volume (J/m^3).



Energy in Robots

- Energy for **motor functions**
- Energy for **non-motor functions** (sensing, control, processing, communication, ...)
- [No energy for self heating: robots can – and should – be “cold-blooded”. Heating is a useless side effect]

Energy in Robots: goals

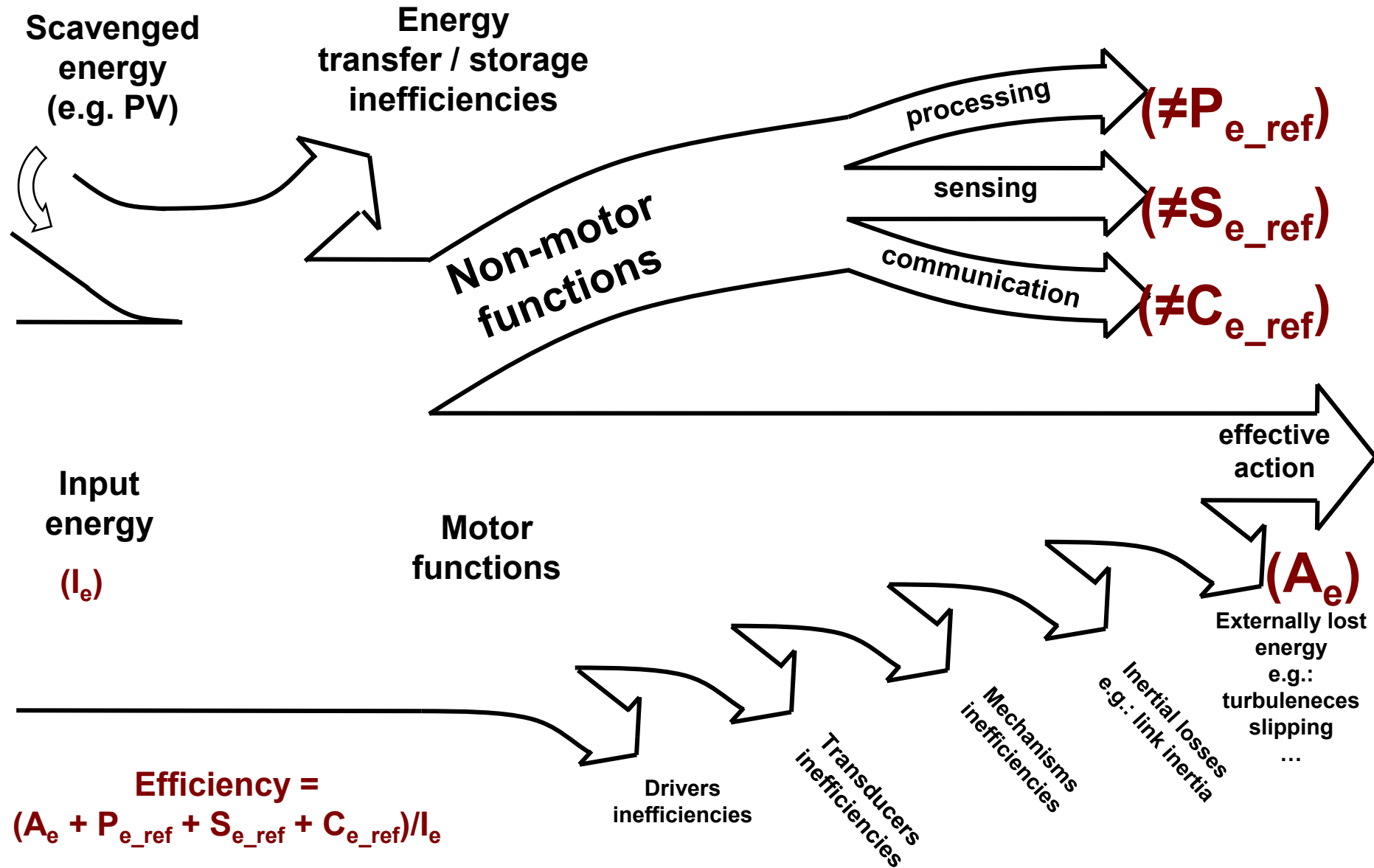
Goals

- Long **autonomy**
- High **performance** (average and instantaneous)
- **Sustainability**

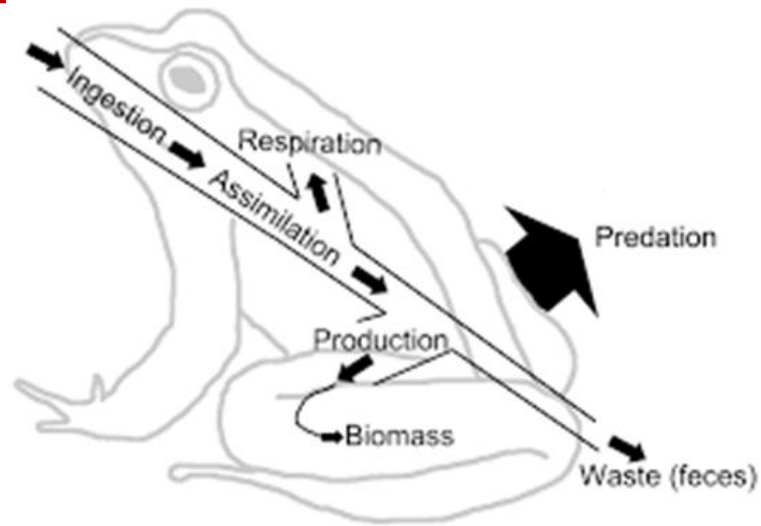
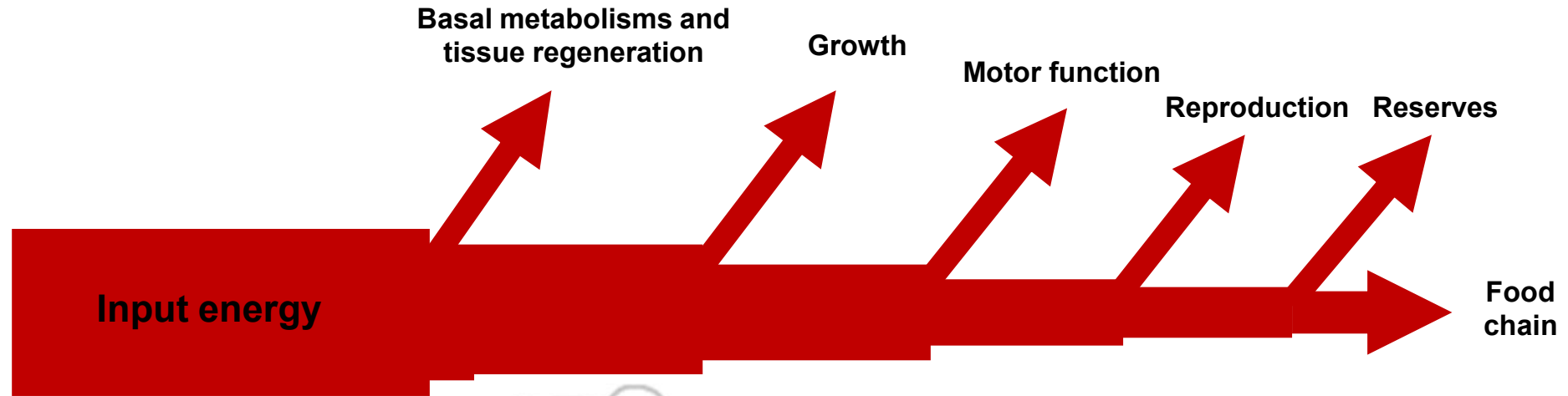
Key Enablers

- **Efficiency** of motor components (mechanisms and actuators), including minimization of energy conversions (e.g.: not only electrical energy as primary source, but also mechanical or chemical)
- **Low-power (ultra low power) electronics**
- High capacity in **on-board storage** of energy
- Energy **harvesting** and **feeding** (organic and inorganic) both in unstructured environments and through energy stations
- Sustainable energy and **easily available** materials

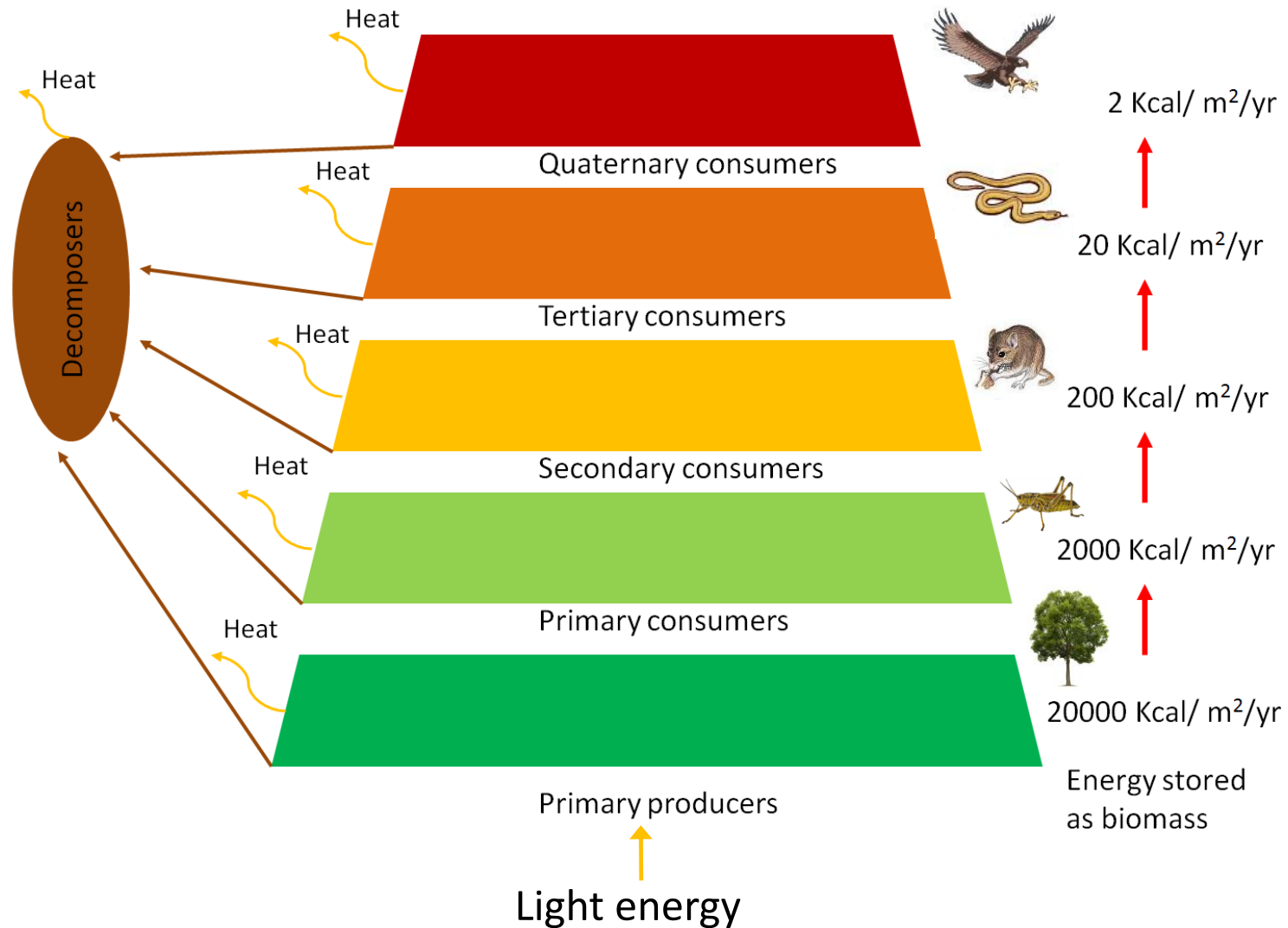
Defining the field



Defining the field



Energy flow in the environment



Energy transformation in living organisms

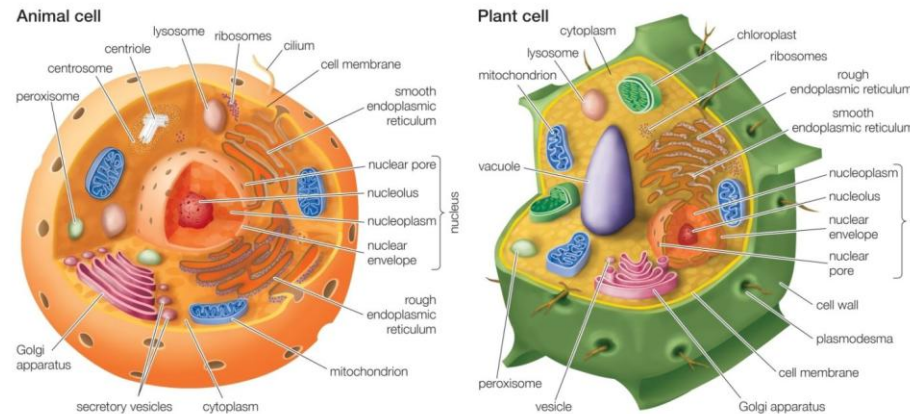
Metabolisms:

- conversion of food/fuel to energy to run cellular processes;
- conversion of food/fuel to building blocks for proteins, lipids, nucleic acids, and some carbohydrates;
- elimination of nitrogen wastes;

Catabolisms



breaking down of organic matter for example, the breaking down of glucose to pyruvate, by cellular respiration



Anabolism



building up of components (e.g. photosynthesis)

Energy transformation in living organisms

In an ecosystem, energy is stored as biomass starting with the plants

Ecosystem production ($\text{g m}^{-2}\text{y}^{-1}$)



Production $P = B/t$



Productivity $p = P/B_1$

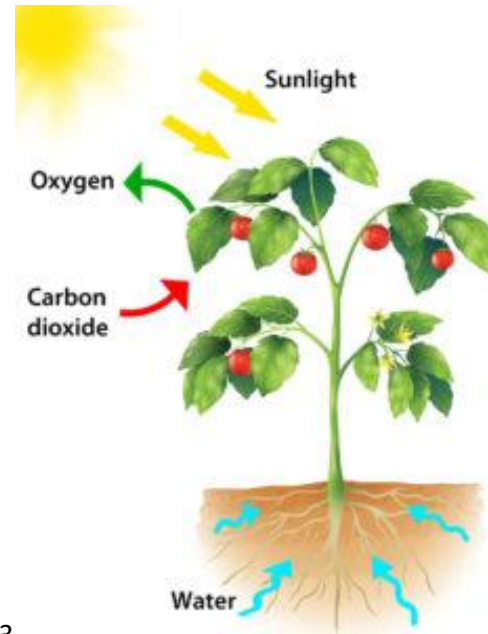


Energy transformation in living organisms

Primary Production

synthesis of organic compounds from atmospheric or aqueous carbon dioxide

generally using light as its source of energy



Gross

Overall biosphere gross production = 101×10^{13}
Mcal year

Net

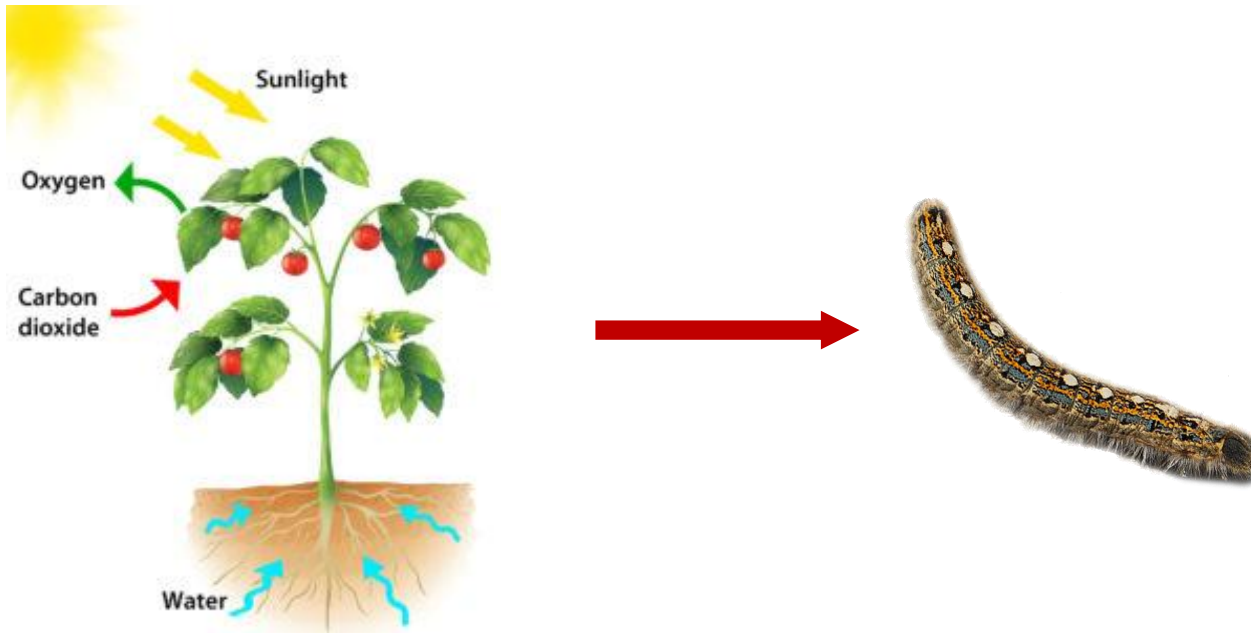
$$NPP = GPP - R_a$$

Secondary Production

Energy transformation in living organisms

Net Community Production

Biomass stored per unit time and space in an ecosystem including all trophic levels of the food chain.

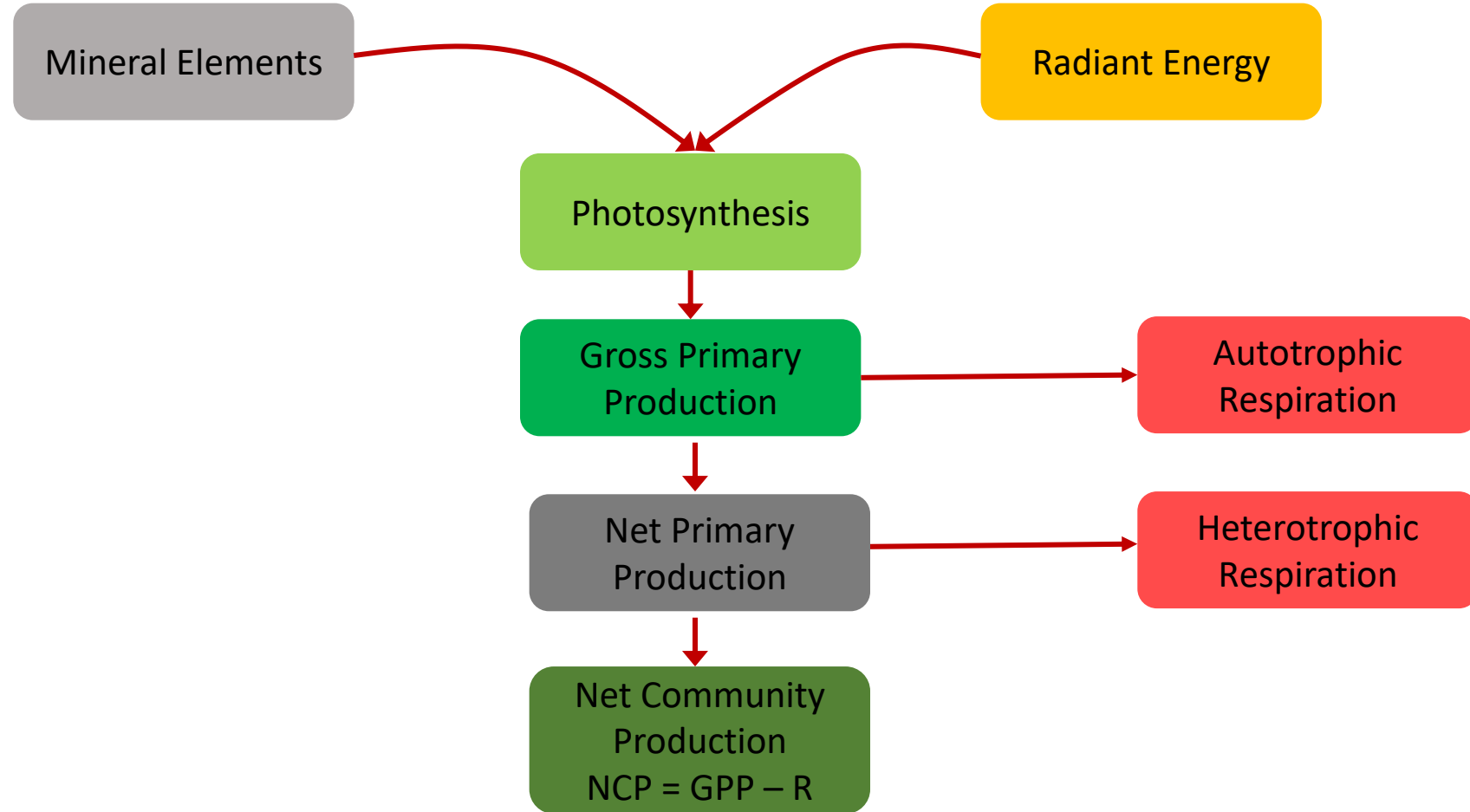


NCP

$$NCP = GPP - R_a - R_e$$

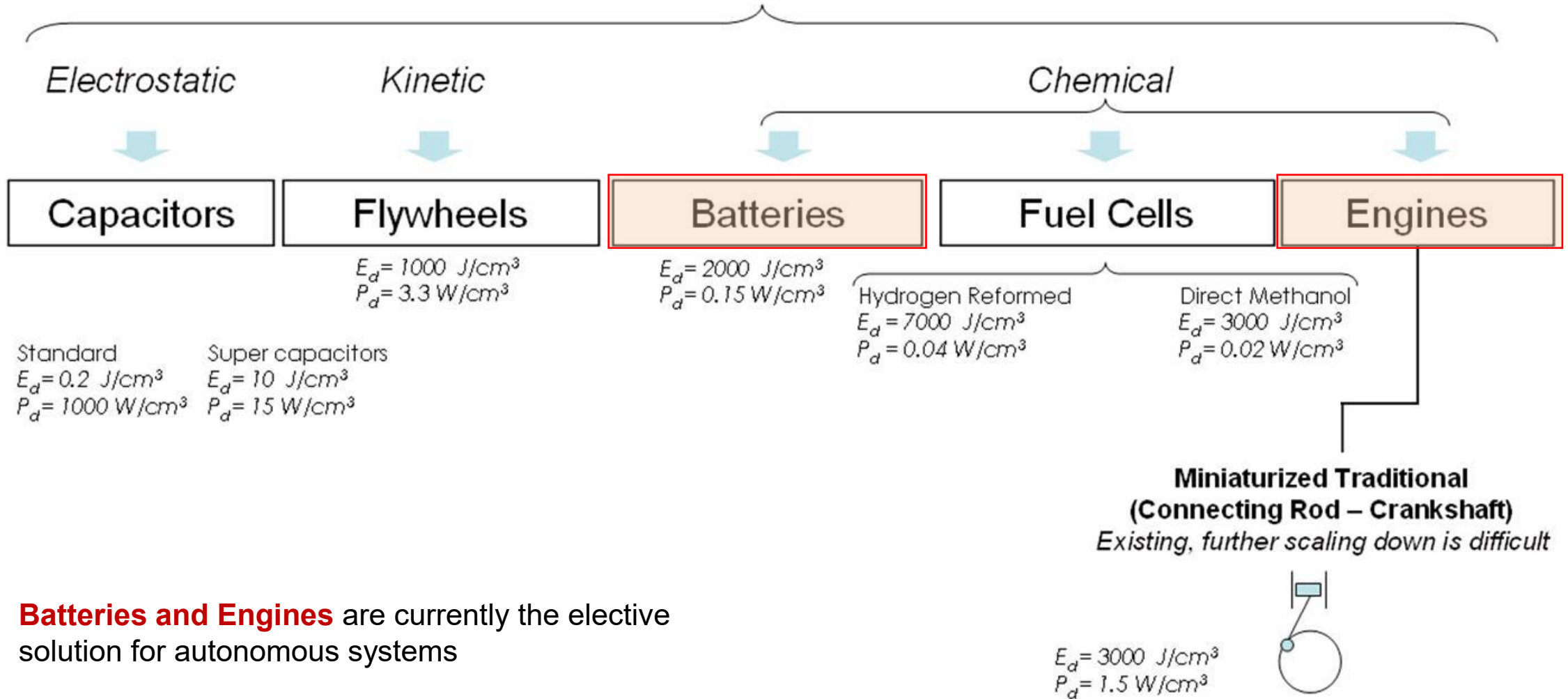
$$NCP = GPP - R$$

Energy transformation in living organisms



High capacity in on-board storage of energy

Type of Stored Energy



Batteries and Engines are currently the elective solution for autonomous systems

Batteries, an overview

Electric **batteries** are devices consisting of one or more **electrochemical cells** with external connections provided to power electrical devices

Convert **chemical energy** directly to **electrical energy**.

The movement of ions within the battery allows current to flow out of the battery to perform work

- **Primary batteries** are used once and discarded (the electrode materials are irreversibly changed during discharge). Higher energy densities than rechargeable batteries
- **Secondary batteries** can be recharged multiple times by restoring the original composition of the electrodes by reverse current

Alkaline Batteries

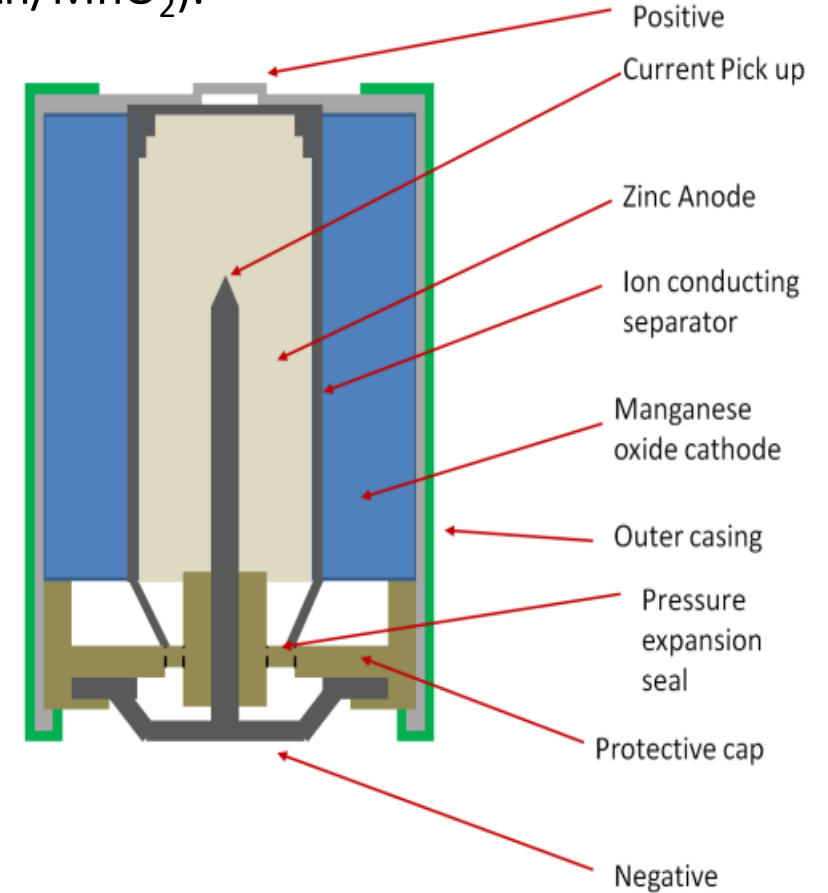
Primary batteries.

(e. g. dependent upon the reaction between **zinc** and **manganese dioxide** Zn/MnO_2).

The **negative electrode** is zinc and the **positive electrode** is manganese dioxide.

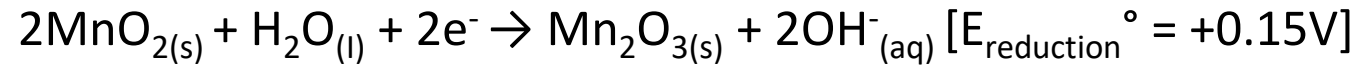
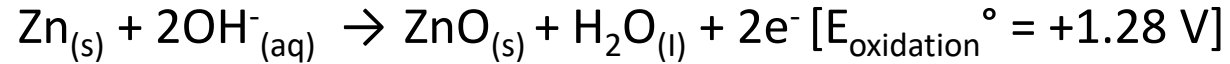
The **alkaline electrolyte** (potassium hydroxide) is not part of the reaction since there are equal amounts of OH^- consumed and produced.

The amount of **current** that an alkaline battery can deliver is roughly proportional to its physical size.

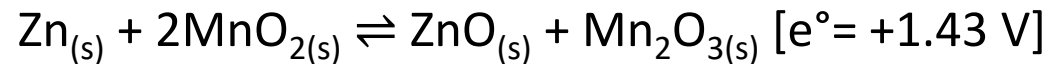


Alkaline Batteries

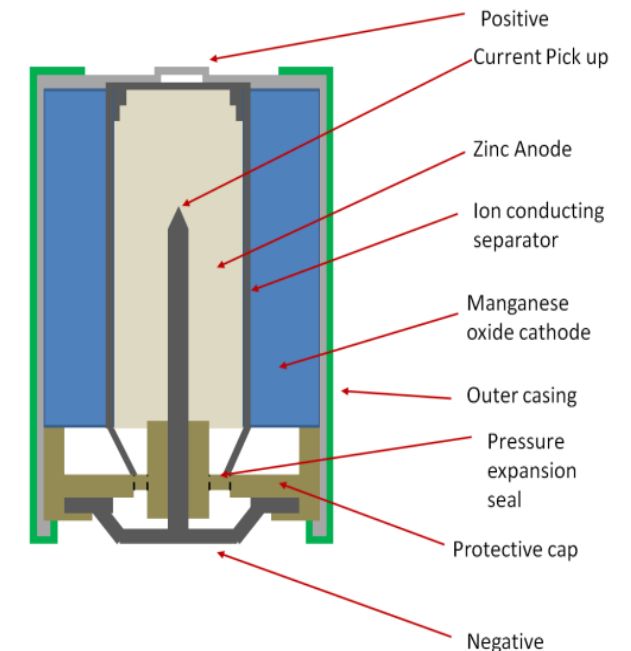
Half-reactions:



Overall reaction:



Nominal cell potential 1.5 – 4.5 – 9.0 V



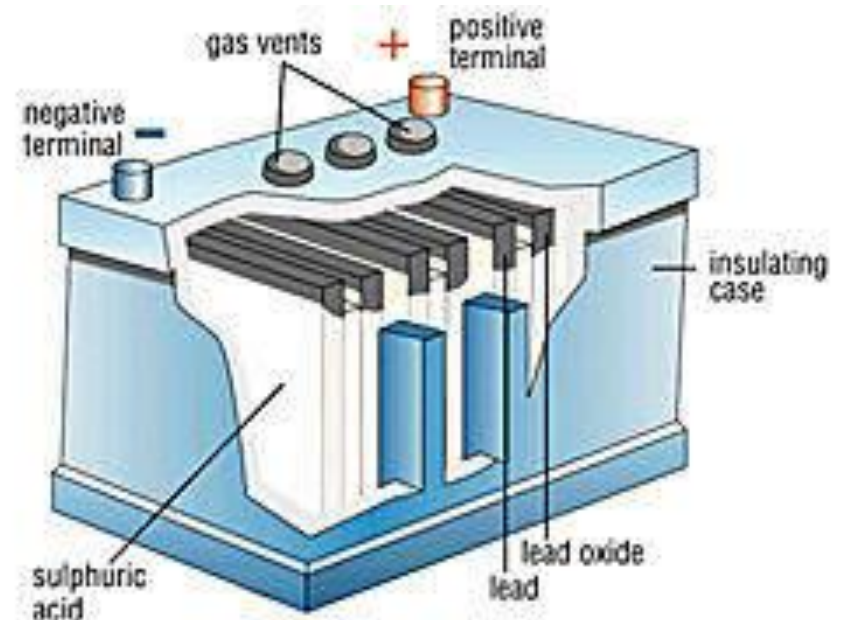
Lead Acid Battery

Lead acid was the first rechargeable battery for commercial use

In the **discharged state** both the positive and negative plates become **lead (II) sulfate** (PbSO_4), and the **electrolyte** loses much of its dissolved **sulfuric acid** and becomes primarily water.

In the **fully charged state**, the negative plate consists of **lead**, and the positive plate **lead dioxide**, with the electrolyte of concentrated **sulfuric acid**.

The **grid structure** of the lead acid battery is made from a lead alloy. Small quantities of other metals (antimony, calcium, tin and selenium) are added to get the mechanical strength of the lead and improve electrical properties.



Lead Acid Battery

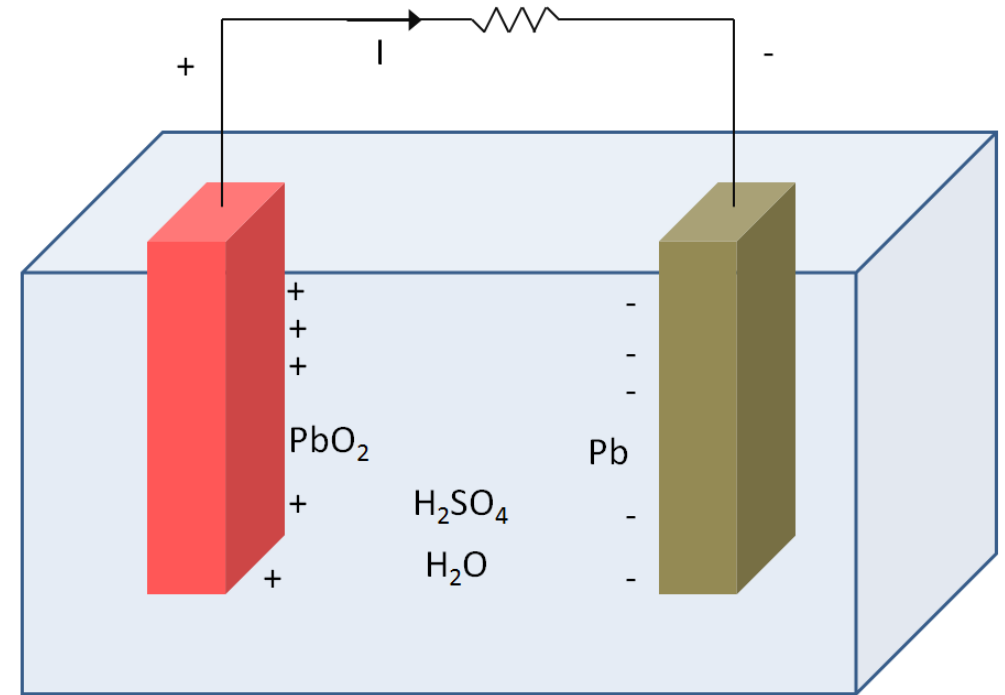
A lead-acid battery's theoretical voltage is **2 V for one cell**.

A full charge takes **14–16 hours**.

The battery must always be stored at **full state-of-charge**.

Low charge causes **sulfation**, a condition that robs the battery of performance.

200 to 300 discharge/charge cycles.

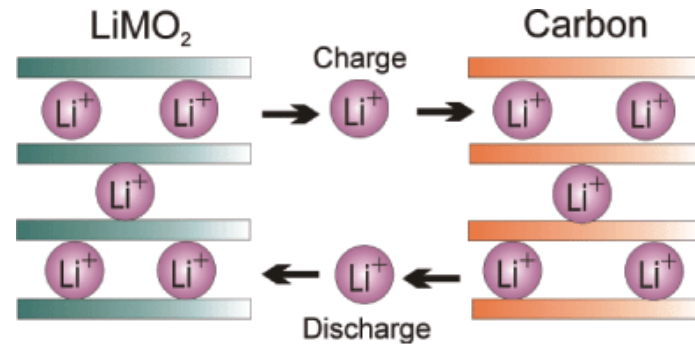


Lithium-ion battery

Lithium: the lightest of all metals, has the greatest electrochemical potential and provides the largest specific energy per weight.

The negative electrode of a conventional lithium-ion cell is made from **carbon**; the positive electrode is a metal **oxide**;

the electrolyte is a **lithium salt** in an **organic solvent**;



Lithium ions move from the **negative electrode** to the **positive electrode** during discharge and back when charging.

Lithium-ion battery

The **voltage**, **energy density**, life, and safety of a lithium-ion battery depend on materials choices.

Pure lithium is highly **reactive** with water to form **lithium hydroxide** and **hydrogen** gas.

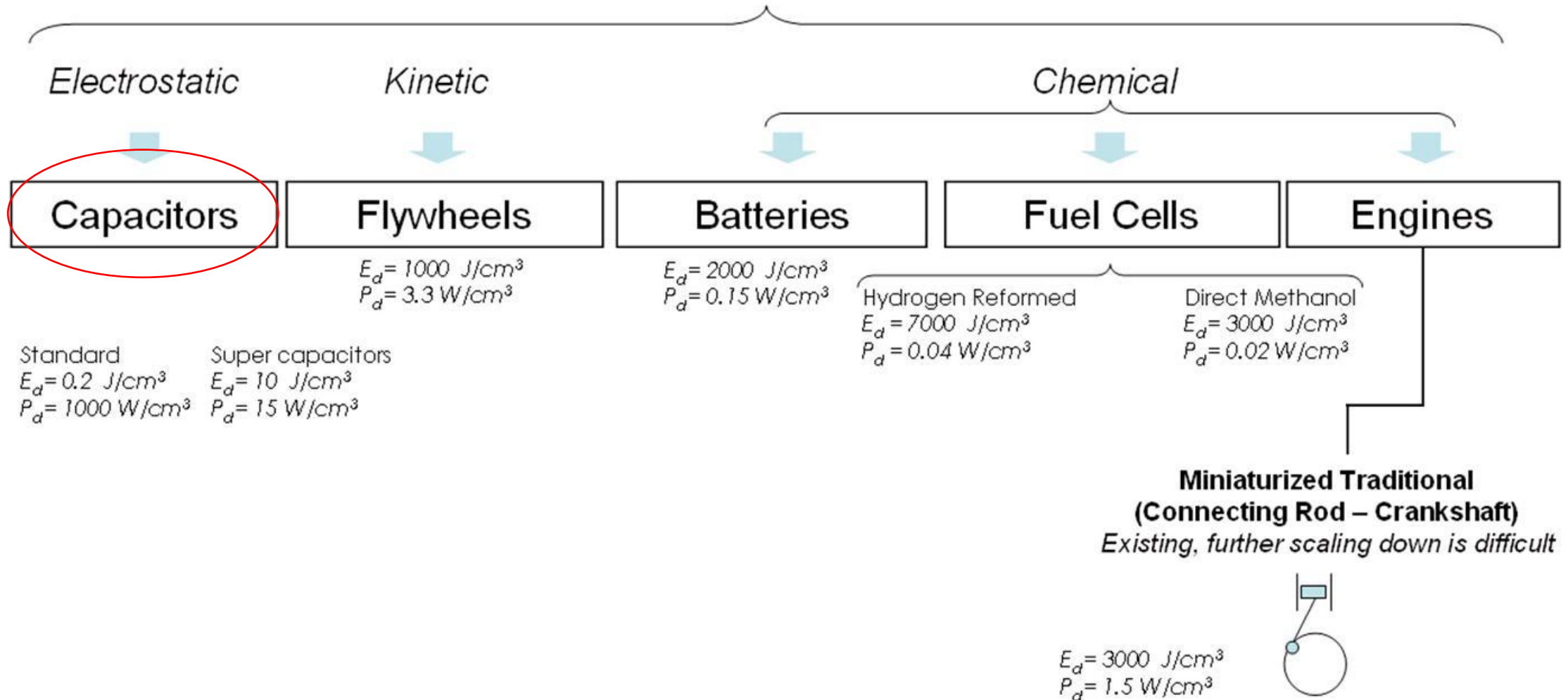
Thus, a **non-aqueous electrolyte** is typically used. A **sealed container** rigidly excludes moisture from the battery pack.

Lithium-ion batteries are more expensive than **NiCd** batteries but operate over a wider temperature range with higher energy densities.

They require a **protective circuit** to limit peak voltage.

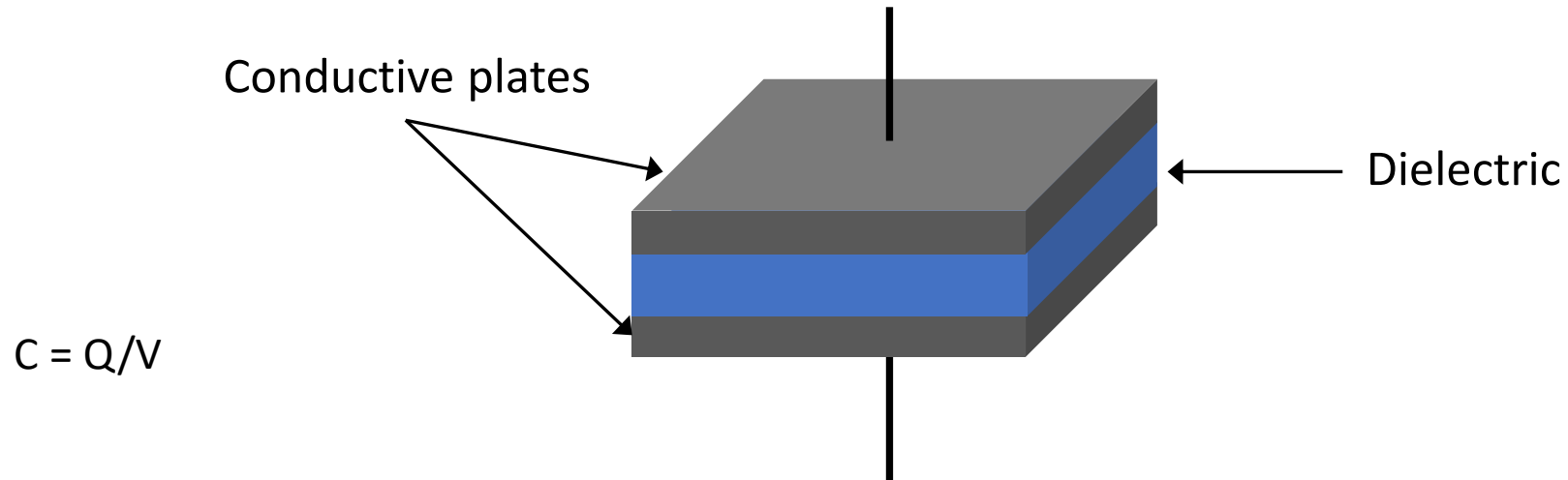
High capacity and efficiency in on-board storage of energy

Type of Stored Energy



Capacitors

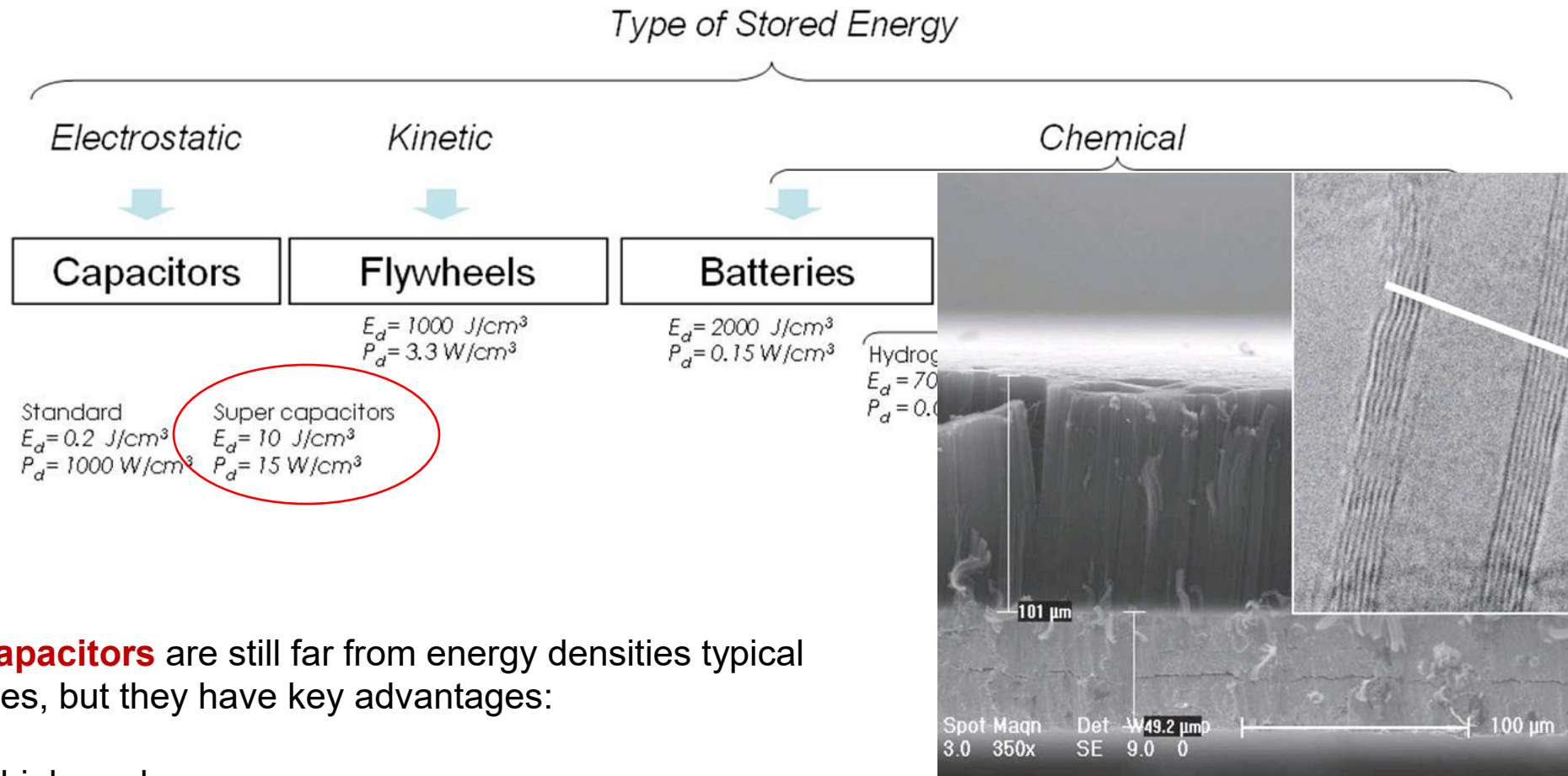
Consisting of two **conductors** separated by a **non-conductive** region (dielectric).



A **capacitor** is specifically designed to provide and enhance the existing **capacitance** between two electrical conductors of a circuit in sufficiently close proximity.

Capacitance increases with the area of the plates and decreases as the distance between the plates increases.

High capacity and efficiency in on-board storage of energy



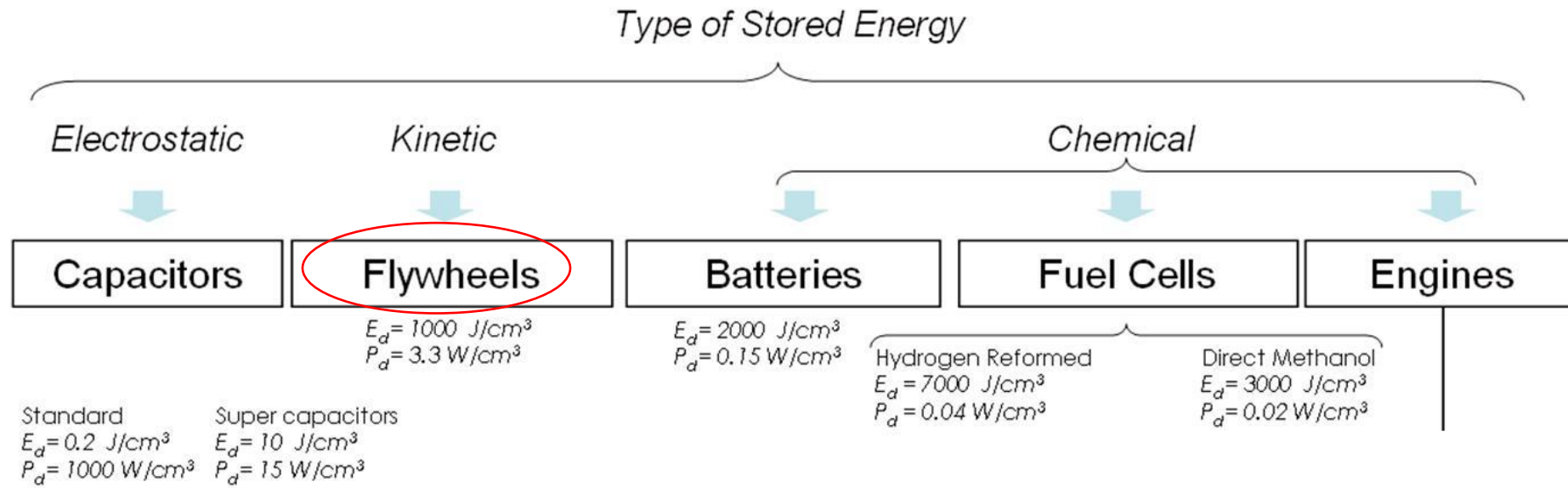
Super-capacitors are still far from energy densities typical of batteries, but they have key advantages:

1. high peak power,
2. high number of recharges
3. no chemical principles (more sustainable materials)

Strategic approach for increasing energy density up to **100x** : nano-structured materials (e.g.: use of nanotubes)

Ultra-capacitors based on Carbon nanotubes are being developed by companies (e.g.: FastCap Systems Co., MA) and they are expected to reach energy densities of 1000 J/cm^3 [MIT Tech. Rev., 2010]

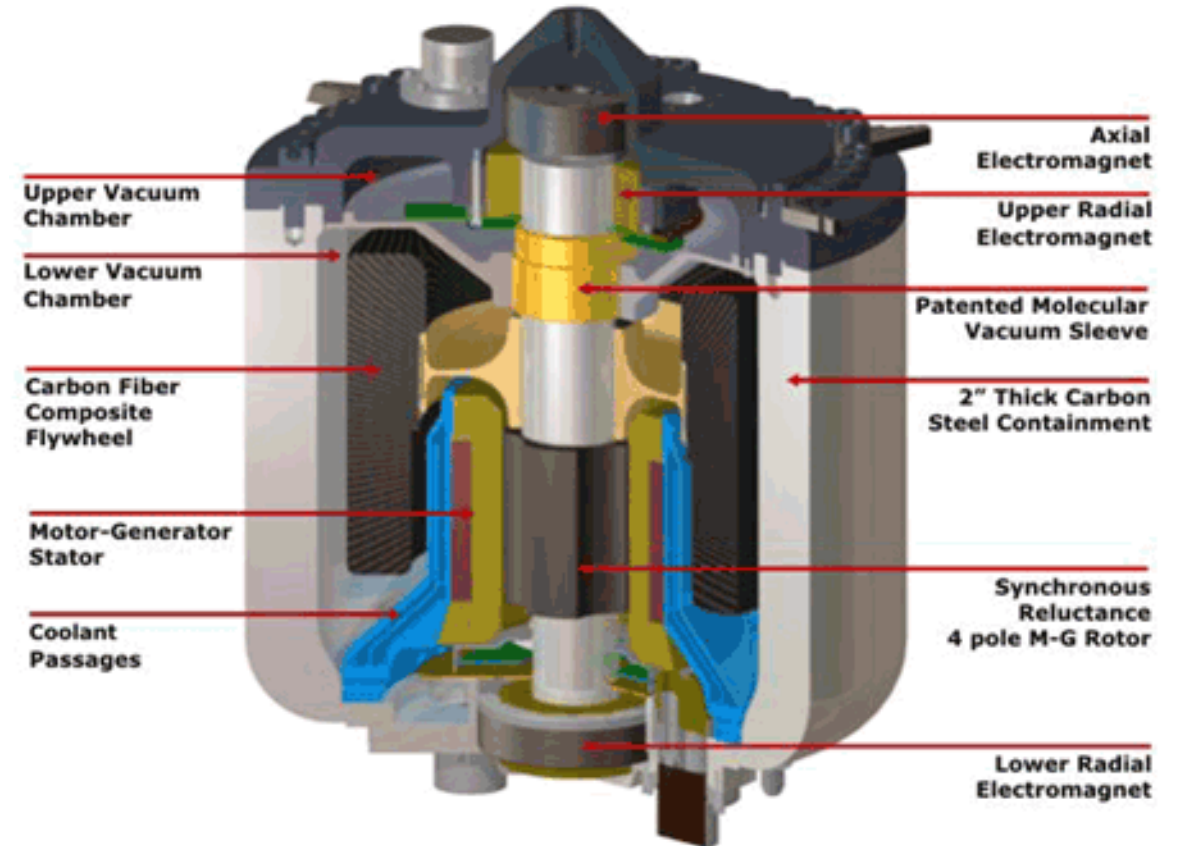
High capacity and efficiency in on-board storage of energy



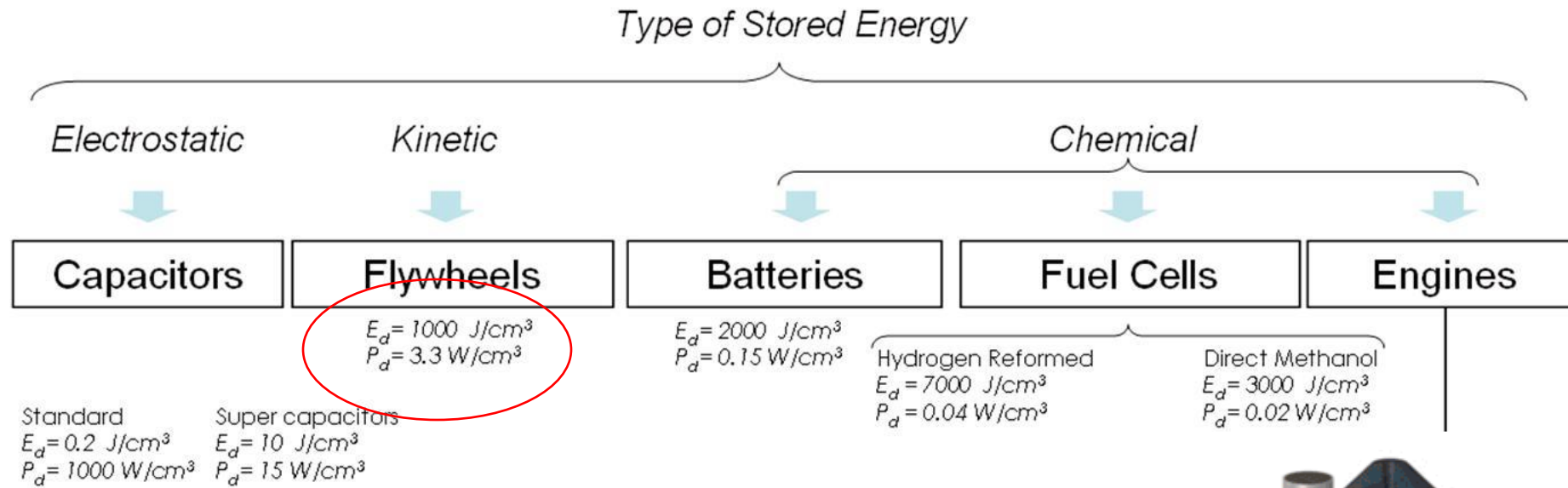
Flywheels

Designed to efficiently store **rotational energy**

A flywheel is supported by **rolling-element bearing** connected to an **electric motor** and enclosed in a **vacuum chamber** to reduce friction and reduce energy loss.



High capacity and efficiency in on-board storage of energy



Flywheels have high potential:

1. high peak power,
2. high energy density
3. Possibility of direct conversion (mech-mech) with high efficiency

The KERS system is an example of the potential of this technology [*"F1 KERS: Flybrid". Racecar Engineering. 2008–11–18*]

